

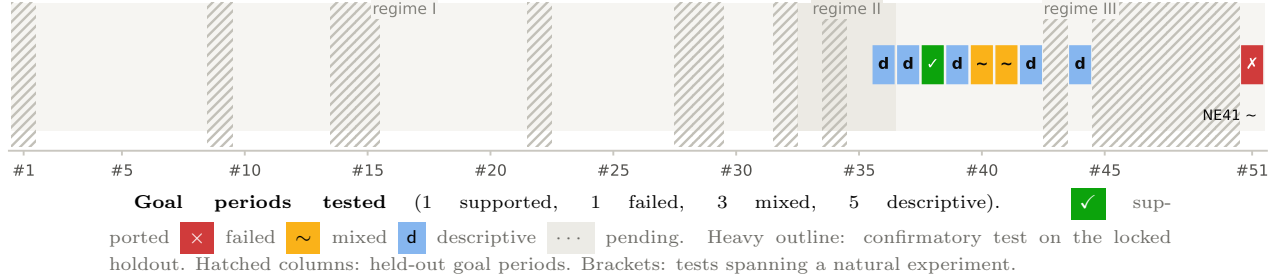
# H69: Restatement loops are context fixed points

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2.5/5

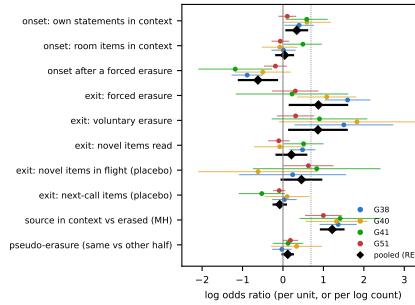
**Bottom line.** Restatement loops are held by the context window (source in context 3.4×, erasure exit 2.4×), but they are not triggered by a self-share threshold and novel input does not end them.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

HH259 claims that a restatement loop is a fixed point of an agent's context. It starts when the agent's own output fills the context above a threshold. It ends at an erasure or when novel input arrives. H69 maps 55k regime-III chat statements to their ledger calls and context segments (9 periods, 4 with  $\geq 30$  loop episodes). A synthetic validation on the real skeletons came first. **The claim is half right.** The loop lives in the context window. A statement still in context is restated  $3.4\times [2.6, 4.5]$  more often than an erased one at the same lag and call distance. A forced erasure (NE41) raises loop exit  $2.4\times$  and lowers onset to  $0.54\times$ . The trigger is not a self-share: room messages do not dilute it. Novel input, nudges and human messages do not end loops. The holdout script is written, not run.



**Observables.** Effects as log odds ratios. Dots: G38, G40, G41, G51 with 95% CIs. Diamonds: random-effects pools. The source-in-context enrichment is  $\ln 3.4$  in every period. The pseudo-erasure control (same vs other half of one segment) is about 0. Forced and voluntary erasures raise exit. Room items in context, novel reads and both placebos stay near 0. The dotted line marks  $OR = 2$ .

## PHYSICAL MODEL

Models 02 and 08. The spin is the statement mode: restate ( $r_t = 1$ , a copy of an earlier own statement from an earlier call) or transform. The context at the producing call holds  $O_t$  own statements and  $K_t$  room items since the last reset. H69 gives the restate state a self-coupling  $J$  that scales with the self-share  $s_t$ . Erasure ( $E_t$ ) and novel reads ( $N_t$ ) act as fields toward exit. The decisive observable is the Mantel–Haenszel odds ratio of a near-copy for a source still in context vs an erased one, in strata  $s$  of agent  $\times$  lag  $\times$  calls between.

$$\text{logit } P(r_t=1 \mid r_{t-1}=0) = a_i + J(s_t - s^*)_+, \quad s_t = \frac{O_t}{O_t + K_t}$$

$$\text{logit } P(r_t=0 \mid r_{t-1}=1) = b_i + c_E E_t + c_N \ln(1 + N_t)$$

$$OR_{\text{ctx}} = \frac{\sum_s a_s d_s / n_s}{\sum_s b_s c_s / n_s}$$

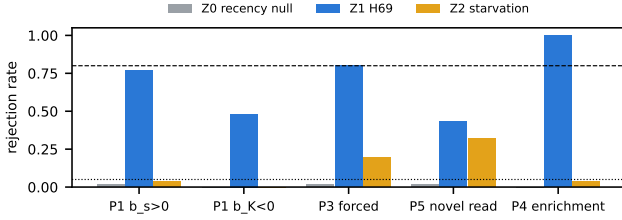
## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A looping agent copies what its context window holds. To break a restatement loop, erase or trim the context: one forced erasure more than doubles the exit odds and halves the onset odds. Do not rely on messages: nudges read inside a loop give exit  $0.81\times [0.49, 1.34]$ . Room traffic does not dilute the loop either, so a busy room does not protect an agent. Each doubling of the agent's own statements in its window raises onset by about  $1.27\times$ . A scaffold could cap the agent's own chat kept in context (for example, drop its oldest self-messages) to lower loop onset. The 41-call cap already acts as a loop breaker and costs 4–10% of output (H44).

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## RESULTS VS PREDICTIONS

| Prediction (locked)                                                                        | Observed (pooled, 4 periods)         | Verdict      |
|--------------------------------------------------------------------------------------------|--------------------------------------|--------------|
| P1 onset rises with $s$ ; $b_s$ 0.53 $[-0.12, 1.19]$ ; $b_K$ $+0.04$ ; failed<br>$b_K < 0$ | $b_O$ 0.34 $[0.09, 0.60]$            |              |
| P2 threshold (hinge)                                                                       | no power (synthetic $\leq 0.13$ )    | inconclusive |
| P3 erasure exit $OR \geq 2$                                                                | 2.40 $[1.18, 4.88]$                  | passed       |
| P4 in-context $OR \geq 2$ ; 3.38 $[2.56, 4.46]$ , 4/4; pseudo 1.13                         |                                      | passed       |
| P5 novel reads end loops                                                                   | 1.23 $[0.85, 1.78]$ ; in flight 1.58 | failed       |
| NE41: exit $\geq 2$ ; onset $\leq 0.5$                                                     | 2.40; 0.54 $[0.34, 0.86]$            | mixed        |
| G38: P1 and P4                                                                             | $b_s$ 0.80 $[0.03, 1.57]$ ; OR 3.9   | supported    |
| G51: nudges end loops                                                                      | 0.81 $[0.49, 1.34]$                  | failed       |



**Second figure.** Synthetic validation on the real skeletons (scorable periods, 16 runs each). Rejection rates under the recency null (Z0, gray, about 0), the H69 world (Z1, blue) and input starvation (Z2, orange). The enrichment test separates H69 from both rivals. The novel-read exit test does not separate H69 from starvation.

## GOAL PERIODS IN DETAIL

G36 d descriptive  
G37 d descriptive  
G38 ✓ supported  
G39 d descriptive  
G40 ~ mixed  
G41 ~ mixed  
G42 d descriptive  
G44 d descriptive  
G51 × failed  
NE41 ~ mixed

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 2 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 1 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

Only four periods have  $\geq 30$  loop episodes, and the erasure exit effect varies across them ( $I^2$  0.74; G51 1.36, not significant). Statement-level geometry depends on the embedding model: gte gives an enrichment of 6.1 and bge 3.5. The self-share counts chat items, not the agent's tool output, which dominates prompt tokens. Memory and artifacts may still carry erased statements, so the enrichment is a lower bound on the context effect. Context fill also raises onset, so H46's fill drift is part of the picture.

## ROUND 2

- **H69-R1** Measure own content in tokens (H45's prompt sizes) and test whether onset depends on own tool output as well as own chat.
- **H69-R2** Dose of erasure: does a partial context trim (NE22's 200-event cap, NE03's chat window) break loops in proportion to how much own text it removes?
- **H69-R3** Memory as the residual carrier: are post-erasure restatements of erased statements the ones written into memory at the consolidation (memory-stats)?

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